

FDA Citizen Petition Analysis

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INTRODUCTION

Background

- The FDA Citizen Petition process enables individuals, organizations and industry stakeholders to request changes to health policy or drug approval decisions
- Process criticized for long, inconsistent response times, with many petitions decided far past the statutory 180-day deadline
- Process remains understudied because FDA correspondence is highly heterogeneous, complex, and difficult to convert into a structured dataset suitable for systematic analysis.

Objective

- Predict FDA expected response times using petition data and NLP-based analysis

METHODS

Data Cleaning

- Combined two FDA-published datasets containing Petitions and responses from 2000–2024, merging by petition ID
- Standardized submission & response dates, FDA centers, submitter types
- Designed NLP preprocessing pipeline (lowercasing, tokenization, punctuation removal, stop-word filtering, and lemmatization)

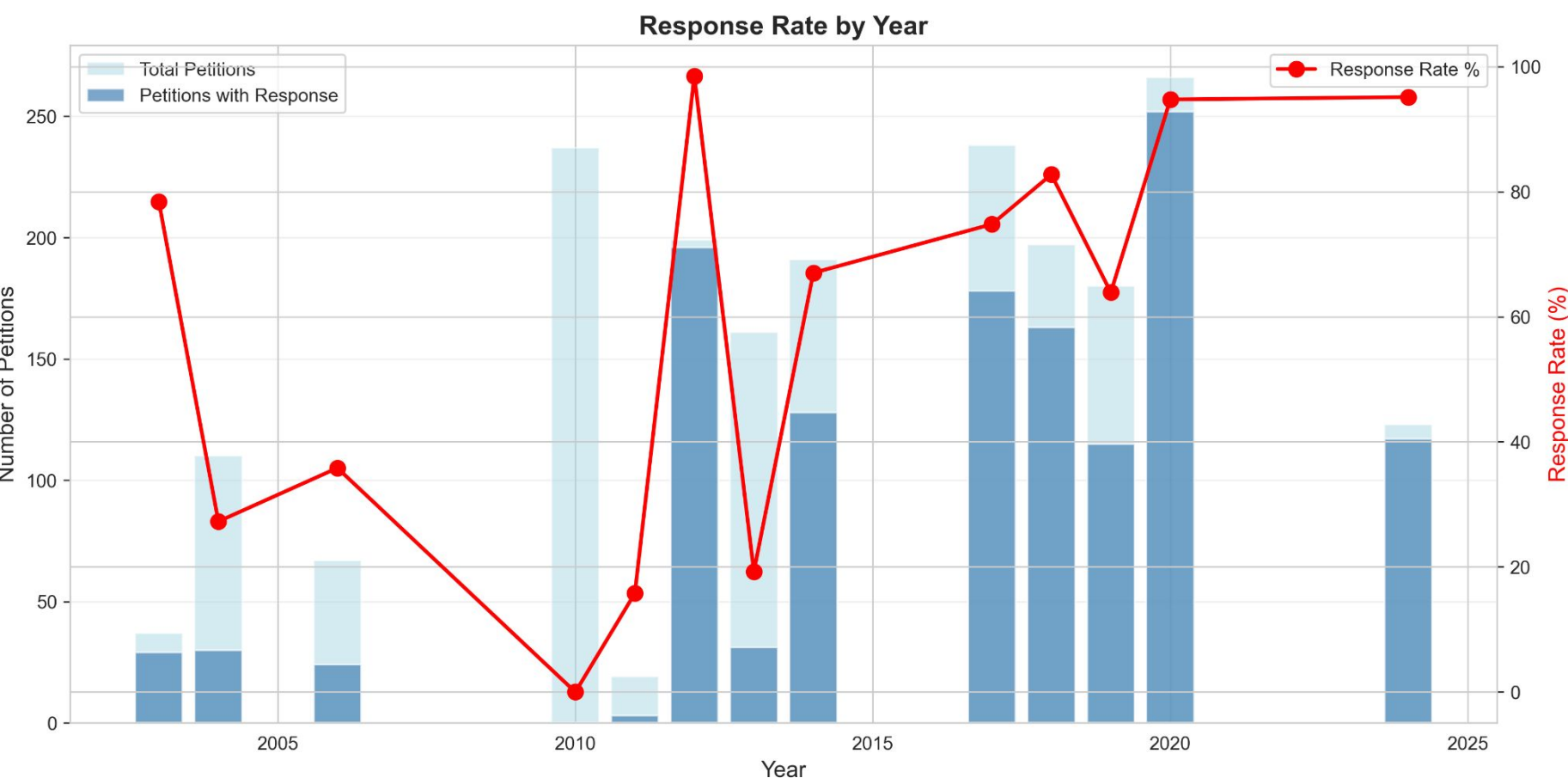


Figure 1: Response rates by year showing how petition volume (bars) and the percentage of petitions receiving a response (red line) vary from 2000 to 2024. The figure highlights both the overall growth in petition submissions and fluctuations in the proportion receiving timely responses.

Feature Engineering

- Response Timing: time-to-response, 180-day delay flag, and censoring for unresolved petitions
- Submitter & Metadata: submitter type, FDA center, petition length, and filing year
- Text Features: TF-IDF vectors, NMF topic proportions, and fine-tuned BERT embeddings

Analytical Approach

- Identified petition themes using TF-IDF + NMF topic modeling
- Predicted response outcomes with BERT and TF-IDF baseline classifiers
- Modeled time-to-response using Cox Proportional Hazards survival analysis

RESULTS

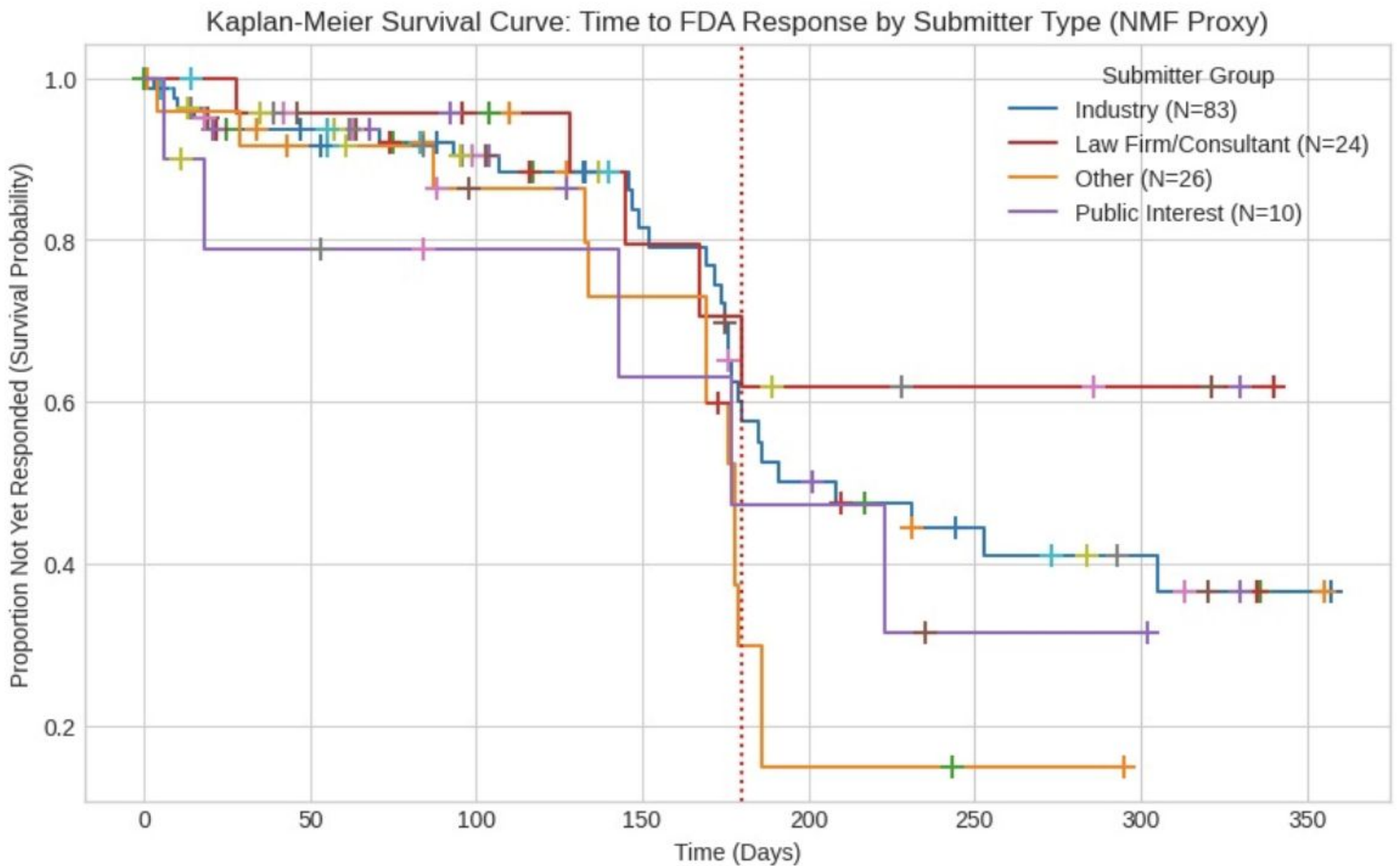


Figure 2: Kaplan–Meier survival curves showing time to FDA response across submitter groups. Industry petitions receive responses more quickly than public-interest or other petitioners. The median response time occurs around 180 days, illustrating how many petitions exceed the statutory deadline.

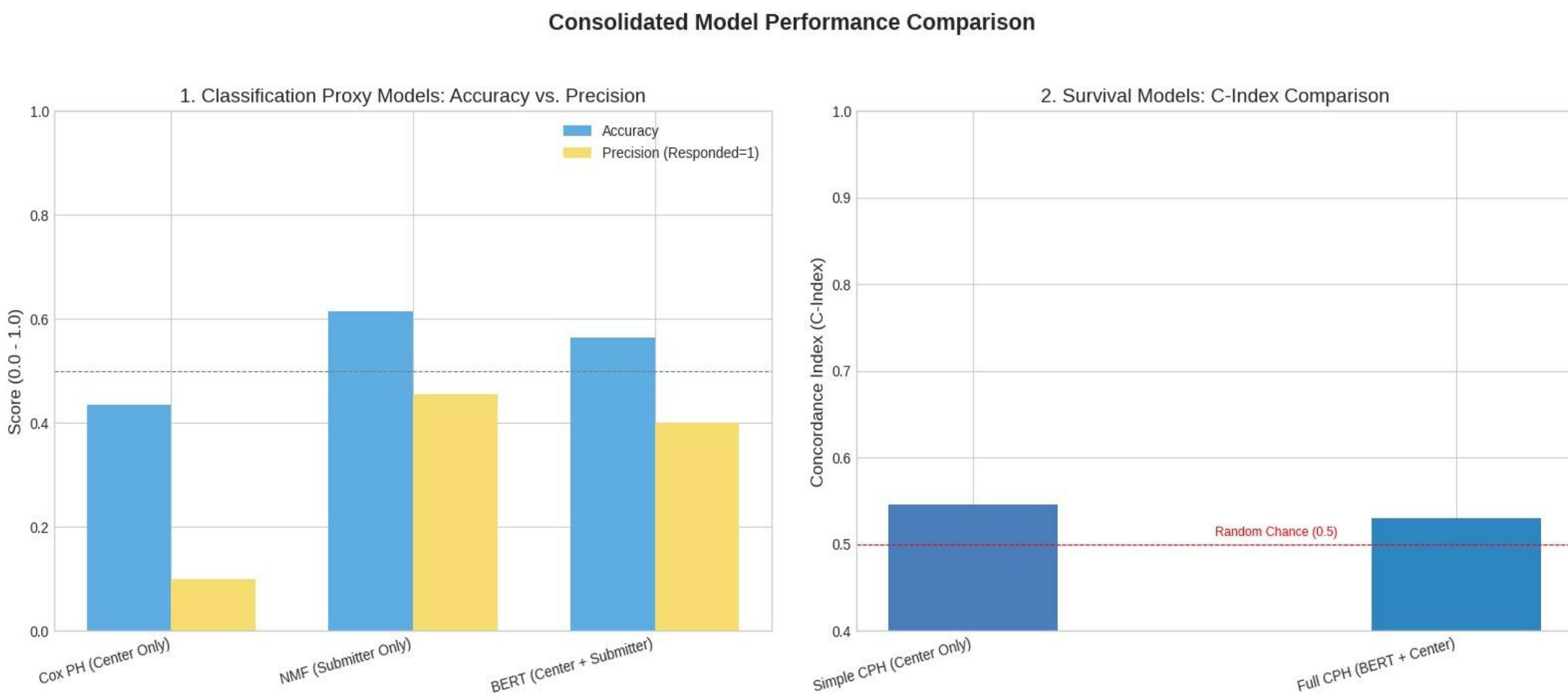
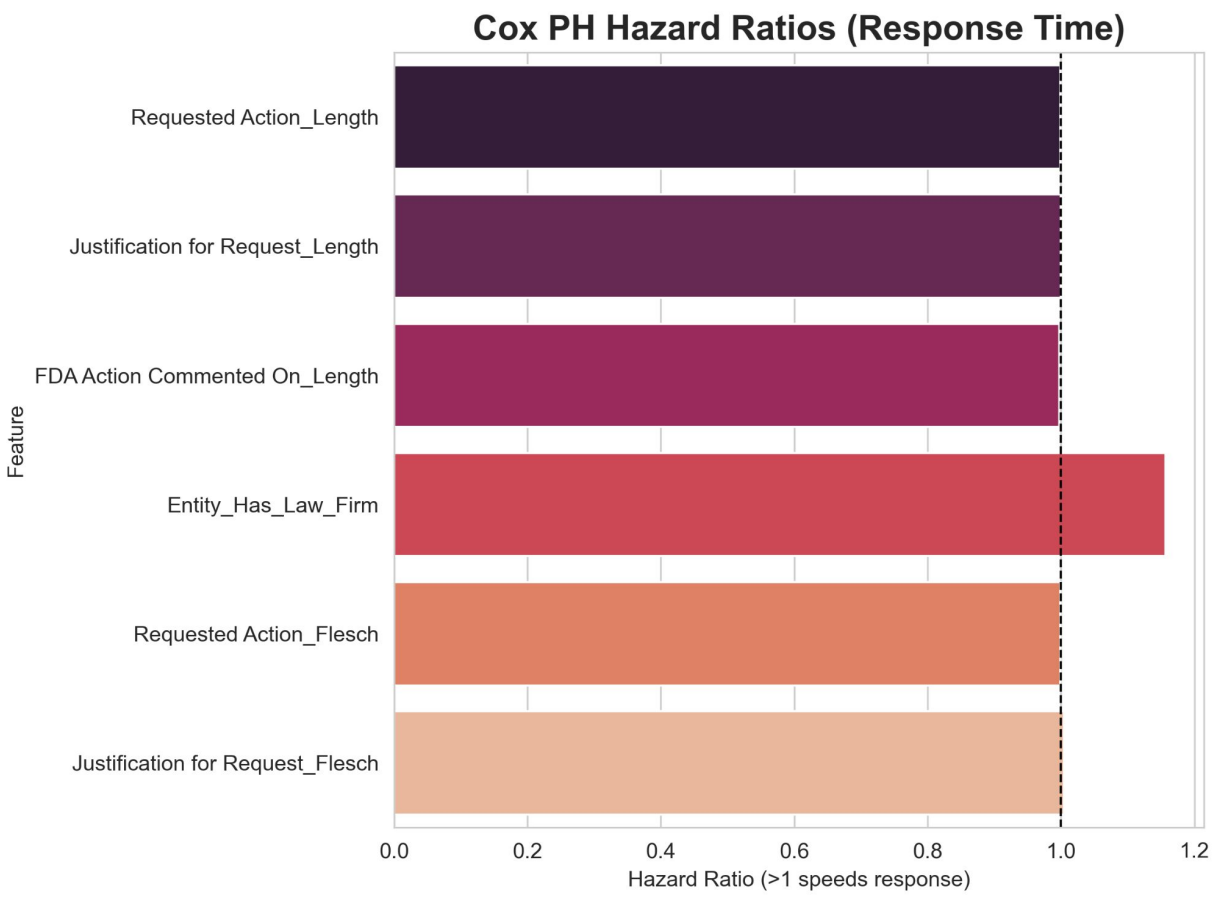


Figure 3: Performance of classification proxy models predicting whether a petition receives a response. The figure shows that models using text and metadata features can meaningfully distinguish between petitions that do and do not receive an FDA response.

Figure 4: Model performance for predicting when a petition receives a response. The C-index scores for both Cox models are only slightly above random, indicating that while some petition attributes influence timing, predicting exact response time remains challenging.

Figure 5: Cox PH hazard ratios showing how petition attributes influence response time. Law firm–filed petitions are associated with faster responses ($HR > 1$), while longer or harder-to-read petitions are linked to slower responses ($HR < 1$).



Key Findings

- Petition volumes fluctuate across years but are consistently dominated by industry.
- FDA responses cluster near the 180-day mark, with long tails of delayed decisions.
- Submitter type strongly shapes response speed, and the Cox model shows that petition characteristics matter too; Classification models effectively predict whether a petition receives a response, but survival models perform only slightly above random.